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Evaluation of Radiation Dose Reduction Strategies in Whole-Body PET/CT imaging

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1. INSTRUCTIONS

Introduction:

Whole-body PET/CT is essential in adult medicine, especially in cancer, where it gives unsurpassed metabolic and anatomical information for diagnosis, staging, and therapy monitoring. 18F-fluorodeoxyglucose (^{18}F-FDG) PET combined with CT can precisely locate hypermetabolic lesions, influencing clinical decision-making in lymphoma, lung, and colorectal malignancies. PET/CT is increasingly used in cardiology, neurology, and infectious disorders, proving its flexibility in diagnostic imaging. Evaluation of adult patient exposure during PET/CT whole body procedure.

Materials and Methods:

The Almana Hospital, Eastern Province, Dammam, Saudi Arabia, PET/CT research center conducted the study. One of the nation's major referral hospitals and cancer centers. The patient's exposure was determined using the precise PET/CT (Siemens Biograph mCT Flow) system and radiation dose.

Results:

The patient's exposure was determined using the precise PET/CT (Siemens Biograph mCT Flow) system and radiation dose. The table size is 78 cm, the rotation time is 0.5 s, and the weight capacity is 227 kg. CT machines output 140 kVp and 800 mA max. Results: 4395 whole-body PET-CT images were evaluated. Patients averaged 55.95 years old (SD = 14). None of the patient BMI groups exceeded 50%. The mean DLP was 842.1 mGy.cm (SD: 200.6). The mean CTDI was 7.58 mGy (SD: 1.14) . Radionuclide activity averaged 327.3 MBq (SD: 38.9). (Table 1).

Conclusions:

Diagnostic quality may be maintained while following ALARA (As Low As Reasonably Achievable) standards. The current results is comparable with previously published studies (Table 2)

Figure 1.Almana Hospital, Eastern Province, Saudi Arabia

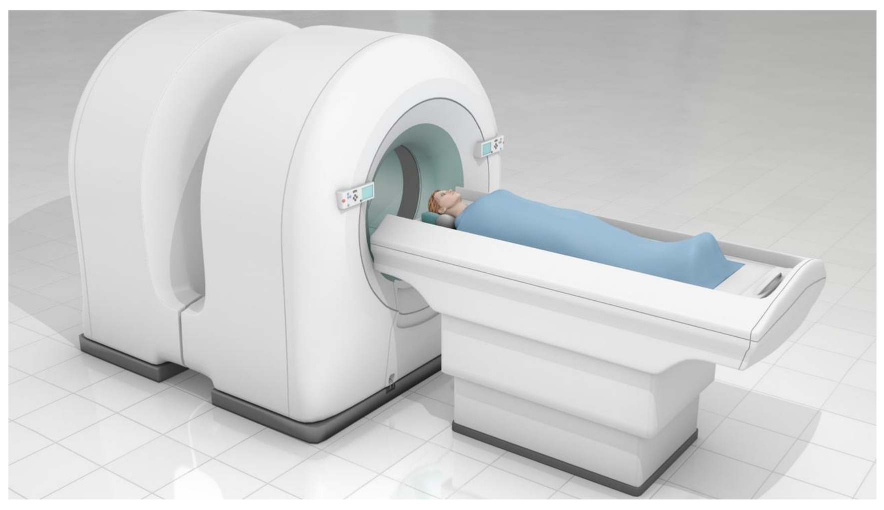

Figure 2.PET CT system

Table 1: PET/CT dose Parameters

Statistic	Activity (MBq)	DLP (mGy.cm)	CTDI (mGy)	Scan Length (mm)	SSDE (mGy)	kVp
Mean	327	842.0	7.58	1119.4	9.0	120
StDiv	38.93	200.64	1.14	254.7	1.48	1.86
Min	233.1	457.23	4.29	152	1.26	80
Max	555	2974.08	21.8	2121	24.2	140

Table 2: Compare our study with the previous study

Author	Activity (MBq)
Current Study	327.29
(Chipiga et al., 2019)	380
(Kaushik et al., 2015)	370

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